

ST MICHAEL, AK, USA

WMO: 720382

Lat: 63.490N

Lon: 162.110W

Elev: 92

StdP: 14.65

Time Zone: -9.00 (NAK)

Period: 06-19

WBAN: 00123

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.6	-18.0	-27.4	1.2	-22.5	-24.1	1.4	-18.1	38.5	14.5	34.7	16.7	6.8	230	0.857

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	67.2	59.0	64.4	57.5	62.5	56.4	60.3	65.3	58.7	63.4	57.0	61.4	10.2	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.8	71.6	63.2	56.0	67.1	60.9	54.5	63.4	59.4	26.7	65.4	25.5	63.4	24.4	61.6	67.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.7	27.5	24.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	30.0	8.4	12.2	10.1	24.7	36.5	46.0	54.8	53.1	45.3	33.4	20.3	13.6
	DBStd	19.72	17.02	14.73	12.60	10.31	9.45	11.86	4.70	4.91	5.74	7.51	9.76	12.76
	HDD50	7669	1289	1058	1238	760	422	177	10	25	157	515	890	1128
	HDD65	12788	1754	1478	1703	1210	884	569	318	369	590	980	1340	1593
	CDD50	357	0	0	0	0	3	58	158	121	17	0	0	0
	CDD65	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH74	9	0	0	0	0	0	2	7	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	10.7	12.4	13.0	11.3	9.4	7.8	8.2	9.9	8.5	9.7	11.9	13.0	13.3
Precipitation	PrecAvg	20.6	1.3	1.3	1.1	0.9	1.0	1.3	2.5	3.3	2.6	2.1	1.7	1.5
	PrecMax	27.0	3.0	4.4	3.1	2.0	2.0	2.2	4.8	6.6	4.9	5.4	3.5	3.1
	PrecMin	13.1	0.3	0.3	0.2	0.3	0.3	0.6	0.4	0.9	1.1	0.6	0.4	0.6
	PrecStd	3.5	0.7	0.8	0.7	0.5	0.4	0.5	1.1	1.1	1.0	1.1	0.9	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.3	38.2	38.2	49.0	59.2	69.8	73.5	69.1	60.9	52.3	40.6	41.9
		MCWB	35.4	34.7	35.0	39.9	51.0	58.2	62.5	60.8	54.7	46.0	37.6	36.7
	2%	DB	36.8	36.1	35.2	43.5	54.3	65.1	68.4	65.6	56.9	48.7	37.4	36.2
		MCWB	34.3	33.4	32.7	38.5	47.1	56.1	59.8	59.2	51.9	44.5	34.7	33.2
	5%	DB	34.2	34.2	32.1	39.7	51.5	61.7	65.6	63.2	55.0	46.0	34.9	33.7
		MCWB	32.0	32.0	30.3	36.0	45.4	54.7	58.8	57.2	50.9	42.8	33.2	31.6
	10%	DB	30.1	31.6	29.5	37.0	48.1	58.7	63.3	60.7	53.7	43.9	32.7	30.2
		MCWB	28.6	29.9	28.1	34.0	43.5	52.9	57.0	56.2	50.1	40.9	31.3	28.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.9	35.2	35.2	41.3	51.3	60.5	64.6	62.2	55.5	46.7	38.7	38.0
		MCDB	38.8	37.5	37.7	46.7	57.9	67.2	71.7	67.0	59.4	50.2	40.3	39.8
	2%	WB	34.4	33.4	32.6	38.7	48.3	57.6	60.9	59.8	52.6	45.2	35.3	33.3
		MCDB	36.9	35.4	34.8	43.7	53.1	63.3	66.0	64.0	55.6	48.8	37.0	35.7
	5%	WB	32.1	32.1	30.5	36.3	45.7	55.2	59.3	58.2	51.2	43.3	33.4	31.5
		MCDB	34.2	34.0	31.9	39.9	50.4	60.4	64.5	62.2	54.6	46.1	34.8	33.3
	10%	WB	28.5	29.9	27.8	34.1	43.7	53.2	57.6	56.6	50.1	41.2	31.3	28.7
		MCDB	29.9	31.7	29.2	36.8	48.0	58.0	62.5	60.3	53.2	44.0	32.7	30.2
Mean Daily Temperature Range	5% DB	MDBR	9.5	10.6	11.6	11.9	12.3	12.6	10.2	9.2	8.3	6.8	7.7	9.3
		MCDBR	11.3	9.8	10.1	13.6	16.0	15.7	15.5	12.9	10.3	8.6	7.6	11.3
		MCWBR	9.8	8.3	8.5	9.7	9.8	8.5	8.8	7.8	6.8	6.5	6.8	10.0
	5% WB	MCDBR	11.3	10.6	9.7	13.9	14.4	14.1	13.9	11.7	9.5	8.8	7.3	10.9
		MCWBR	9.9	9.3	8.2	10.3	9.2	8.3	8.4	7.4	6.5	6.8	6.7	9.9
Clear-Sky Solar Irradiance	taub	0.194	0.265	0.268	0.340	0.362	0.365	0.364	0.384	0.318	0.285	0.206	0.184	
	taud	1.970	2.133	2.236	2.081	2.236	2.385	2.385	2.444	2.369	2.539	2.389	1.985	1.892
	Ebn at Noon	135	210	266	262	268	271	268	249	250	207	122	62	
	Edh at Noon	10	20	27	40	38	33	31	30	21	16	9	4	
All-Sky Solar Radiation	RadAvg	66	273	832	1407	1656	1706	1387	1079	679	295	83	26	
	RadStd	10	37	81	96	161	161	162	88	136	64	36	15	2

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (13 neighbors)	+1.49	N/A	N/A	N/A	+0.86	+0.99	N/A	N/A	+63	N/A

Nomenclature: See separate page